BRACT's

Vishwakarma Institute of Information Technology

Department of Information Technology

Faculty Mentor: Dr. Shalini Wankhade

TITLE: TrustVote: Online Voting using BlockChain

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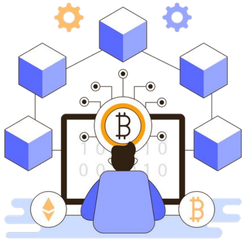
1. Problem Statement

Traditional voting systems often suffer from issues such as voter fraud, lack of transparency, and slow results, which can undermine public trust and participation. A blockchain-based e-voting system offers a solution by providing a secure, transparent, and tamper-proof platform for elections. This project aims to develop a blockchain e-voting system to enhance the integrity and efficiency of the voting process.



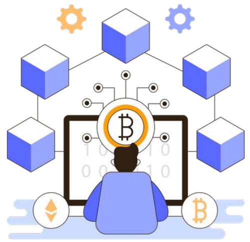
2. Abstract

- The project focuses on designing and implementing a **decentralized voting system** using **blockchain technology** to provide a **secure, transparent, and immutable platform** for elections.
- Utilizing smart contracts on the Ethereum blockchain, the system automates the voting process, ensuring accurate vote counting.
- Developed with **Solidity** for smart contracts, Truffle for Ethereum development, **HTML, CSS, and JavaScript for the frontend**, and **Node.js, Express.js, and MongoDB for the backend**, the system secures users' identities and voting records through encryption and private keys.

3. Introduction

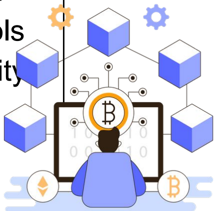
What is Decentralized Voting?

Decentralized voting spreads voting control across many locations instead of one central point. **Secure voting systems** are essential to prevent cheating and ensure fair elections. **Transparent voting** systems allow everyone to verify that votes are counted correctly. **Blockchain** can help by securely recording each vote in a way that can't be changed. This technology ensures that the voting process is open and trustworthy for everyone.

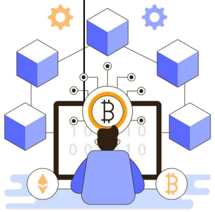


4. Literature Survey

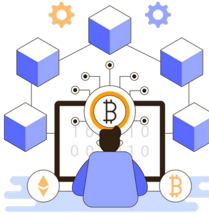
Id	Year,Title,Author	Methodology	Limitation	Comparison to Current Study
[1]	**2020** **A Secure Digital Voting System based on Blockchain Technology** S. S. Bhalerao, et al.	Proposed a secure digital voting system using blockchain technology, emphasizing the system's architecture, security features, and implementation details.	Focused primarily on the technical aspects, with limited discussion on scalability and user accessibility.	This study builds on the concept by addressing additional aspects such as scalability and practical deployment in large-scale elections.
[2]	**2018** **Blockchain-Based E-Voting System** A. Kiayias, et al.	Implemented a blockchain-based e-voting system, focusing on privacy, verifiability, and resistance to tampering.	Lacked practical deployment analysis and did not consider large-scale election scenarios.	The current study compares the implementation strategies and enhances the security protocols for better real-world applicability.



[3]	**June 2016** **Electronic Voting Service Using Block-Chain** Kibin Lee, Joshua I. James, Tekachew Gobena Ejeta, Hyoung Joong Kim	The paper introduces a novel voting system that leverages blockchain technology to enhance security and transparency. It involves several steps: filtering transactions, implementing a consensus protocol, and using a decentralized ledger to record votes. The system incorporates a trusted third party (TTP) for additional validation and employs blockchain for immutability.	The system's effectiveness depends on the accurate functioning of the TTP and the blockchain protocol. Challenges include potential vulnerabilities in the transaction filtering process, maintaining consensus in a decentralized network, and addressing scalability issues. Additionally, the system's dependence on the TTP could introduce points of failure or bottlenecks.	This study Maintains redundant copies of the blockchain data to prevent data loss and ensure continuity in case of an attack or failure.



[4]	**2019** **Blockchain And Finger Print Enabled E-Voting** Aravind , Gokul Raj ,Mohanraj	Proposed system Fingerprint Authentication for Identity Verification due to faster and better than other biometric data.They have used Blockchain Data Storage and Homomorphic encryption. Also the system has two types of users one is Election Officer & another is Booth Manager, Booth Manager System developed with voter’s functionality where voters are going to poll.	The paper mentions that no full implementation of blockchain-based e-voting has occurred for a national election. Although the proposed system aims to reduce costs associated with traditional voting methods, the complexity of implementing this may present significant challenges.	The proposed system aims to decentralize the voting process, potentially increasing security and transparency.

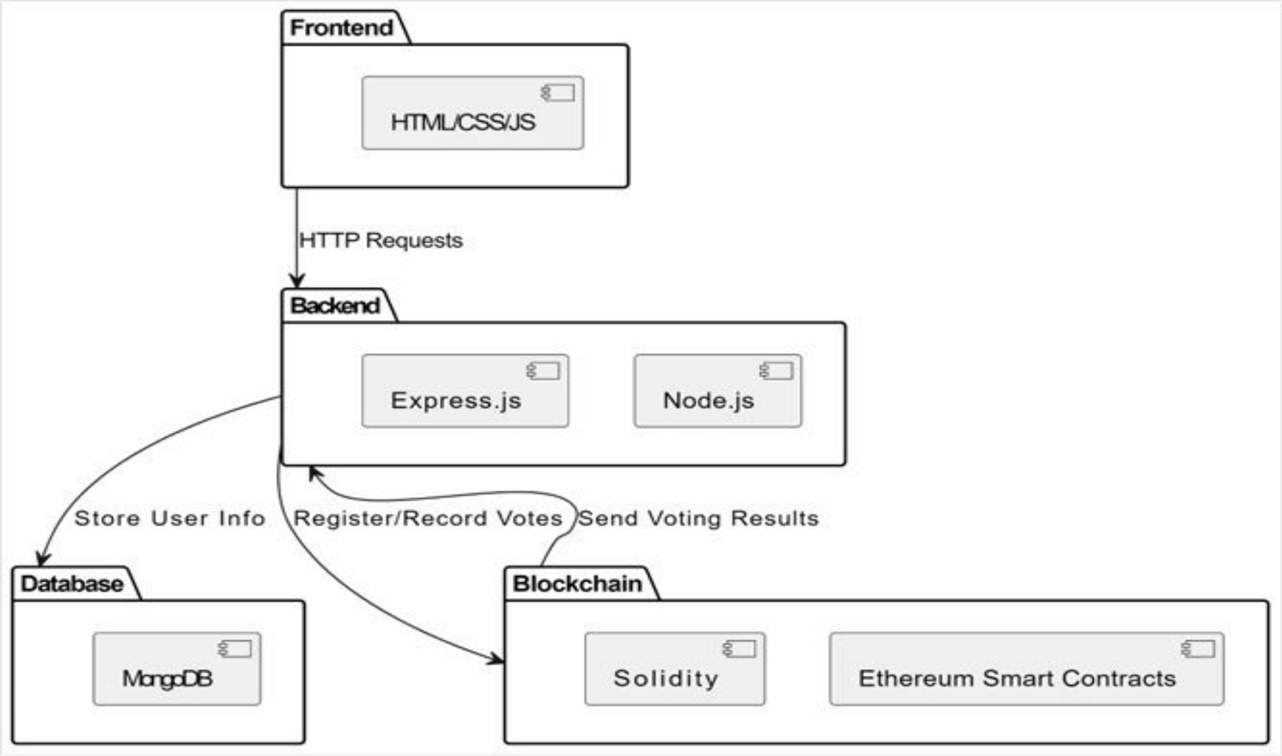


5. System Architecture

- The proposed decentralized online voting system is built using a three-layered architecture, consisting of the frontend, backend, and blockchain. The frontend is developed using HTML, CSS, and JavaScript, providing a user-friendly interface for voter registration, login, and vote casting.
- The backend, developed with Node.js and Express.js, handles server-side operations, including authentication, data validation, and communication with the blockchain. MongoDB is used to store non-sensitive user data such as login credentials.

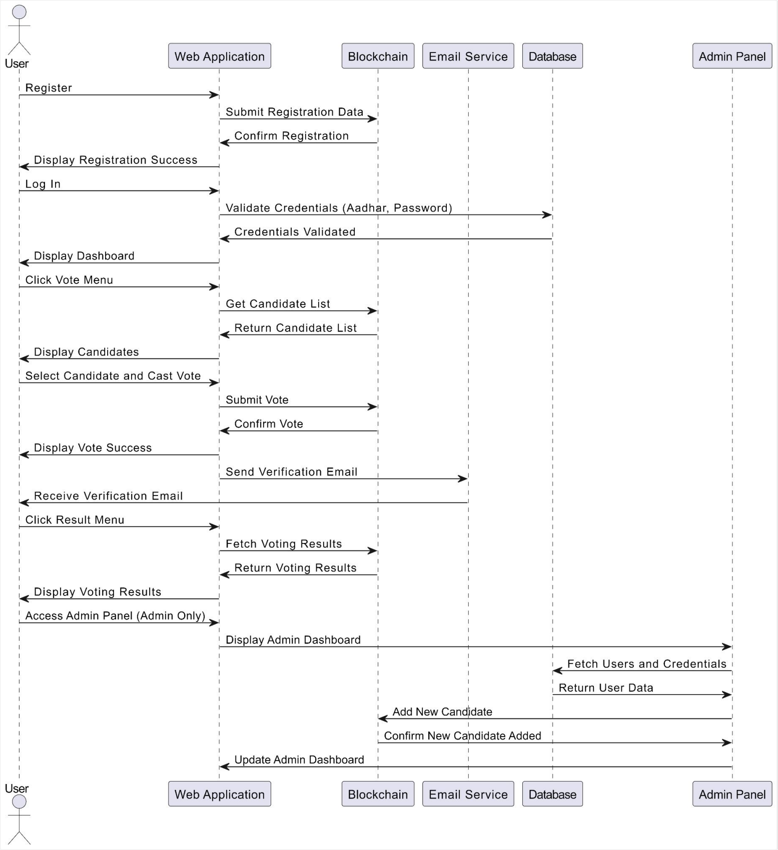
- The core of the system is the Ethereum blockchain, where smart contracts written in Solidity automate critical functions such as voter registration, vote recording, and result tallying.
- Each vote is securely recorded on the blockchain, ensuring immutability, transparency, and preventing tampering. This architecture ensures secure, decentralized management of the voting process while providing a scalable and transparent solution.

5. System Architecture

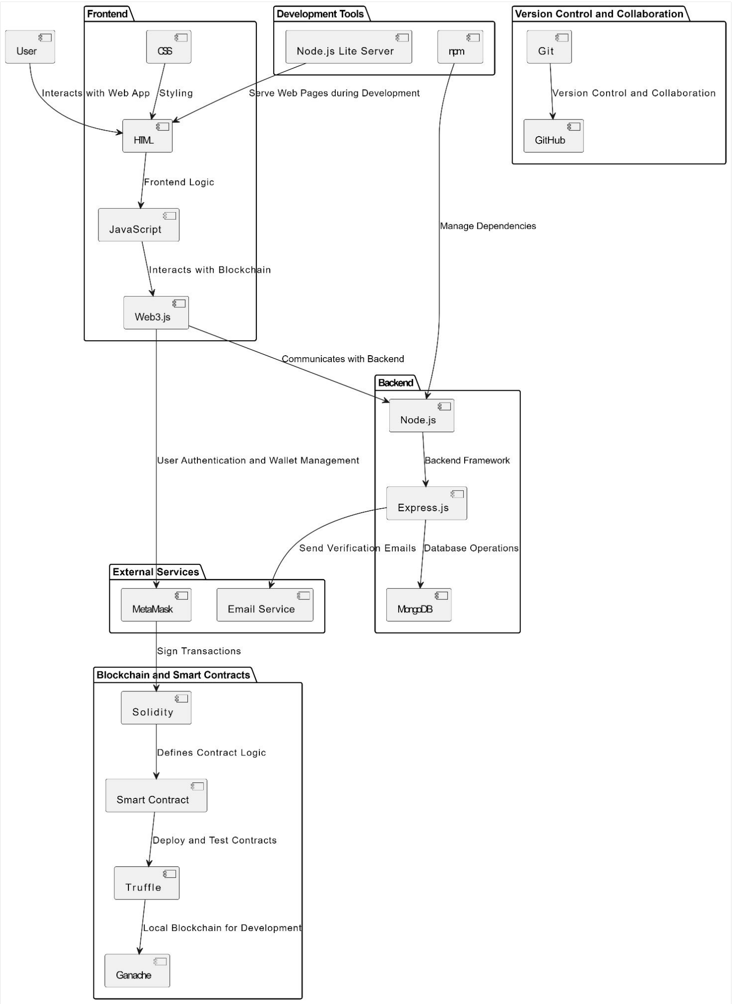


6. UML Diagram

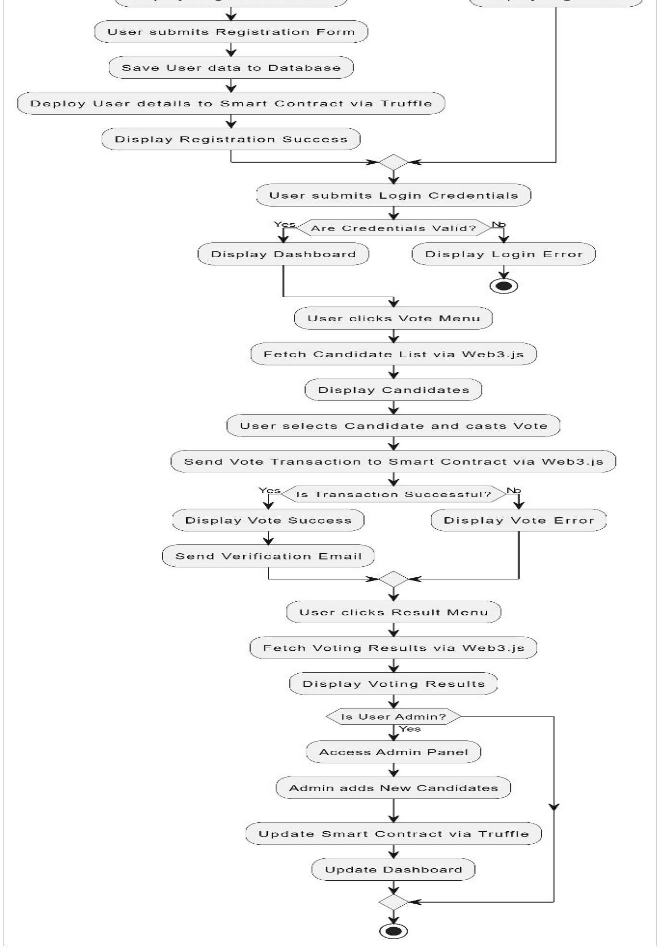
Sequence Diagram



Block Diagram



Flow Diagram



7. Proposed System/Methodology

- In this system the voter/user must first register themselves using a registration form available within the web application and once the registration form is being submitted, an entry is made in the blockchain.

- After the registration, the user can log into the application and be a part of the polling process. The user with valid credentials can log into the system by entering their Aadhar number and password.

- Once the user is logged into their respective account the dashboard contains all the information about voting.

- When users click on the vote menu, they will see the candidates of the respective parties.

7. Proposed System/Methodology

- Users will select one party and cast the vote for that party. Only one vote is casted from each account.

- Once the vote is cast an Email verification will be sent to the registered account and that account is registered into the blockchain.

- The system has a secure login system which prevents people from casting votes on behalf of others.

- By clicking the result on the menu ,the user can see the result on the dashboard.

- Dashboard also contains an admin panel only accessed by the admin; the admin can add new candidates into the system and see the user’s name with its valid credentials.

8. Software Tools/Technologies

Frontend Development:

- **HTML:** Structures web content.
- **CSS:** Styles web pages, controlling layout, colors, and fonts.
- **JavaScript:** Adds dynamic and interactive features to the web.
- **Web3.js:** Facilitates interaction with the Ethereum blockchain.
- **Node.js Lite Server:** Serves web pages with live-reload for development.

Backend Development:

- **Node.js:** JavaScript runtime for building fast, scalable server-side applications.
- **Express.js:** Framework for creating robust web applications and APIs.
- **MongoDB:** NoSQL database for handling large volumes of data.

8. Software Tools/Technologies

Blockchain and Smart Contracts:

- **Solidity:** Language for writing smart contracts on the Ethereum blockchain.
- **Truffle:** Toolset for developing, deploying, and testing Ethereum smart contracts.

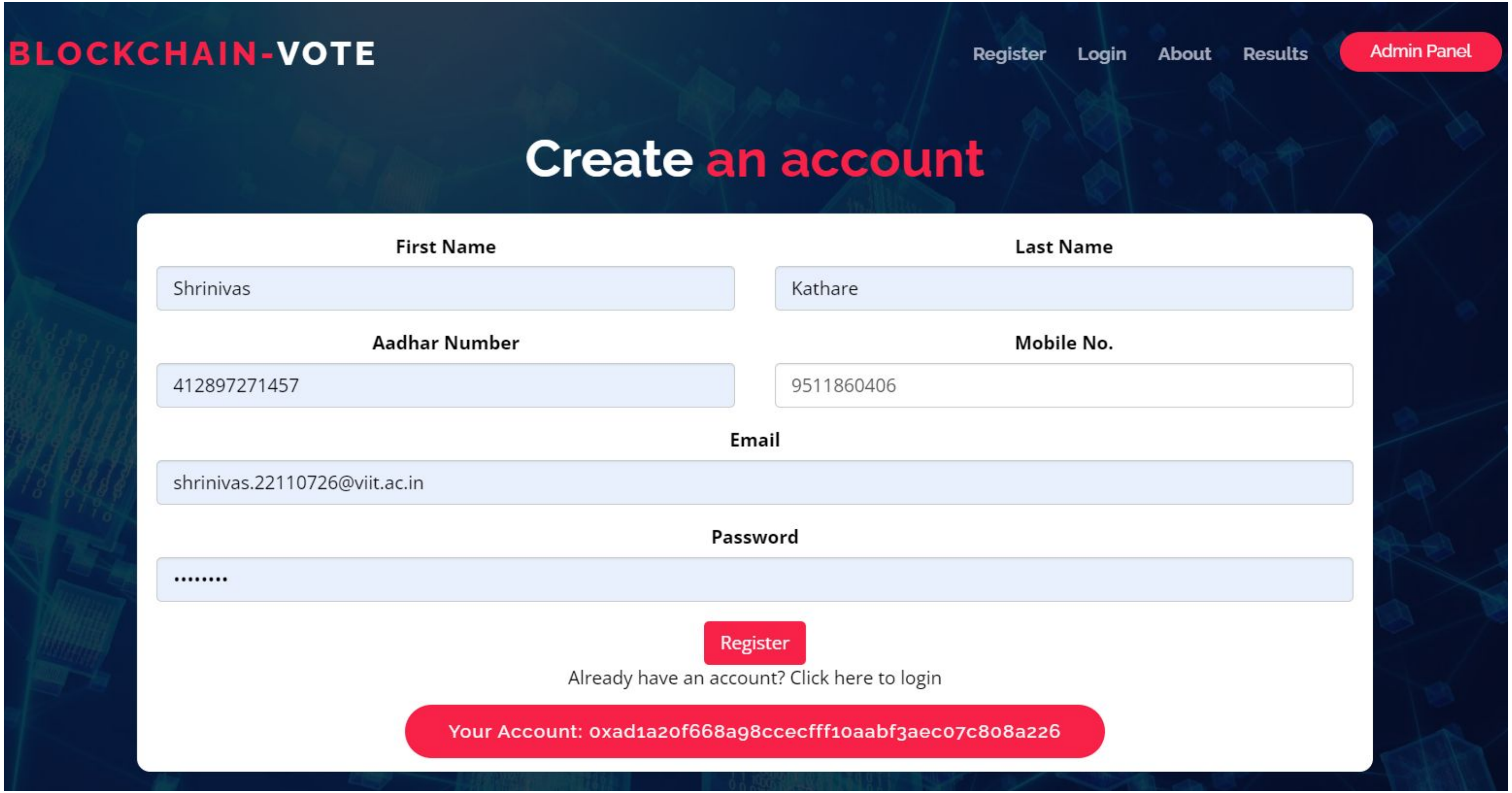
Version Control and Collaboration:

- **Git:** Tracks code changes, enabling collaboration.
- **GitHub:** Hosts Git repositories with additional collaboration tools.

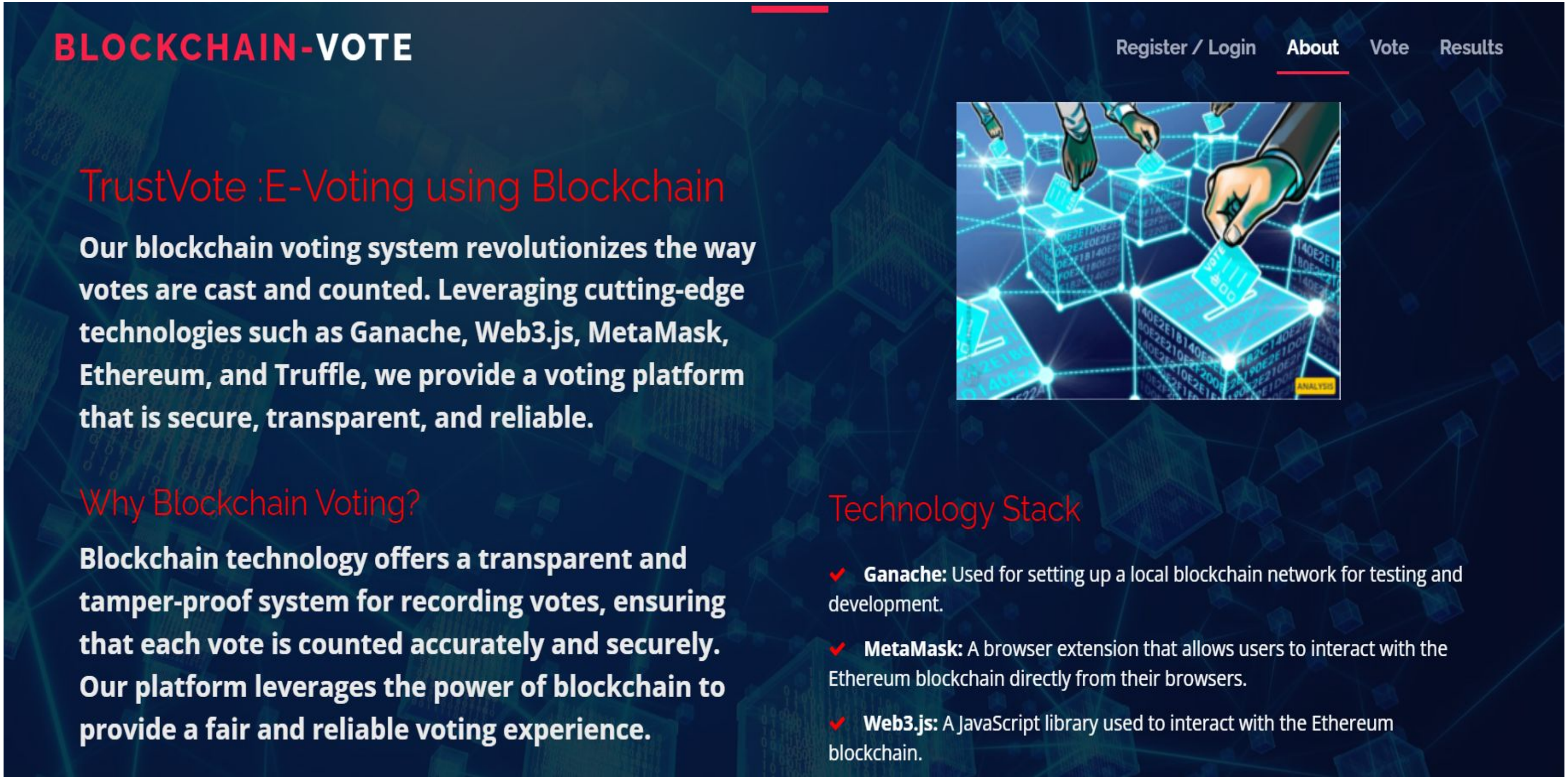
Development and Deployment Tools:

- **Ganache:** Personal blockchain for testing Ethereum contracts.
- **MetaMask:** Browser extension for interacting with Ethereum-based DApps.
- **npm:** Manages project dependencies and packages.

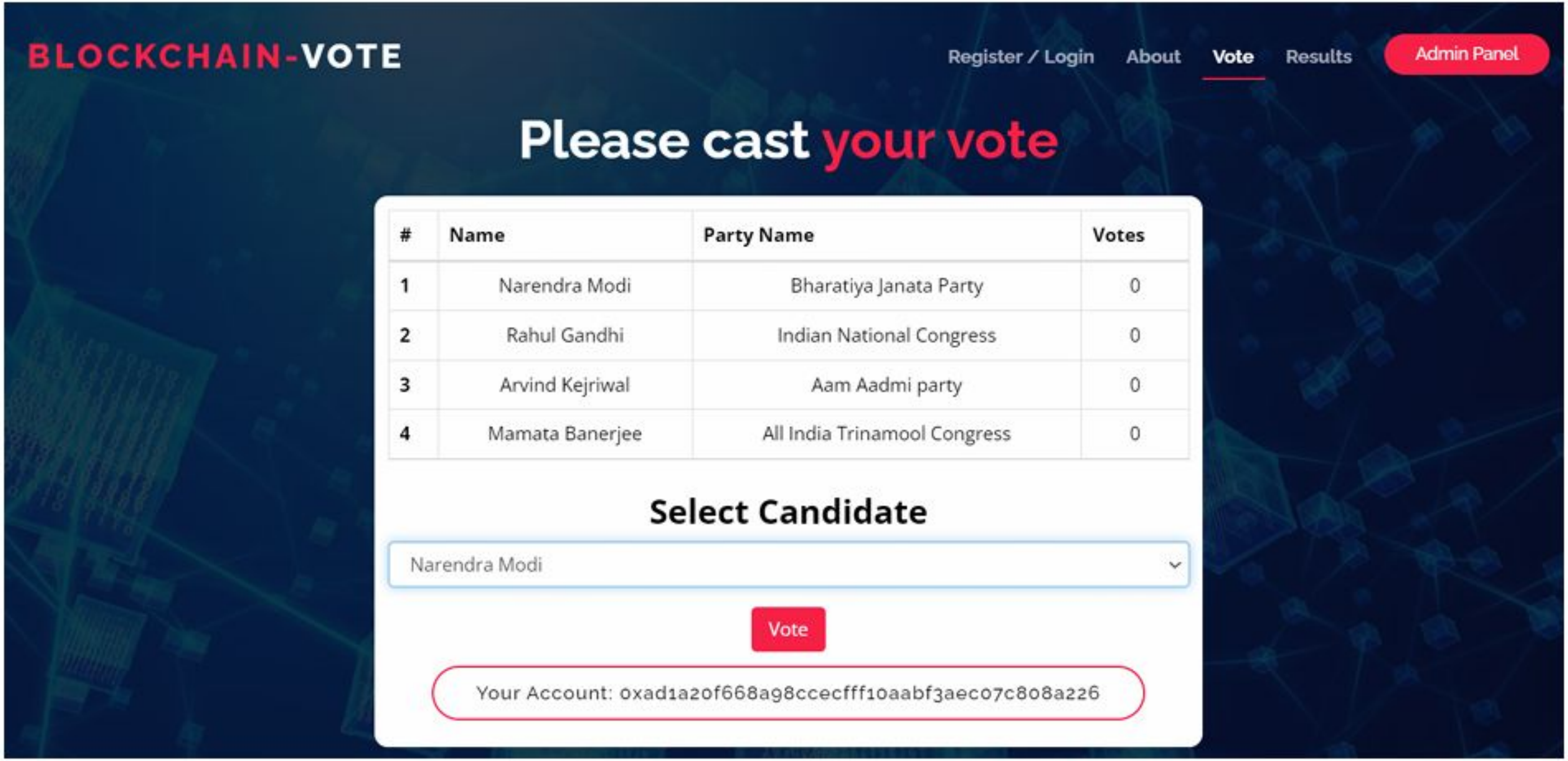
- **Sign-in page**


BLOCKCHAIN-VOTE
Register
Login
About
Results
Admin Panel
Create an account
First Name
Shrinivas
Last Name
Kathare
Aadhar Number
412897271457
Mobile No.
9511860406
Email
shrinivas.22110726@viit.ac.in
Password
••••••••
Register
Already have an account? Click here to login
Your Account: 0xad1a20f668a98ccecfff10aabf3aec07c808a226

- # About


BLOCKCHAIN-VOTE
Register / Login
About
Vote
Results
TrustVote :E-Voting using Blockchain
Our blockchain voting system revolutionizes the way votes are cast and counted. Leveraging cutting-edge technologies such as Ganache, Web3.js, MetaMask, Ethereum, and Truffle, we provide a voting platform that is secure, transparent, and reliable.
Why Blockchain Voting?
Blockchain technology offers a transparent and tamper-proof system for recording votes, ensuring that each vote is counted accurately and securely. Our platform leverages the power of blockchain to provide a fair and reliable voting experience.
Technology Stack
Ganache: Used for setting up a local blockchain network for testing and development.
MetaMask: A browser extension that allows users to interact with the Ethereum blockchain directly from their browsers.
Web3.js: A JavaScript library used to interact with the Ethereum blockchain.

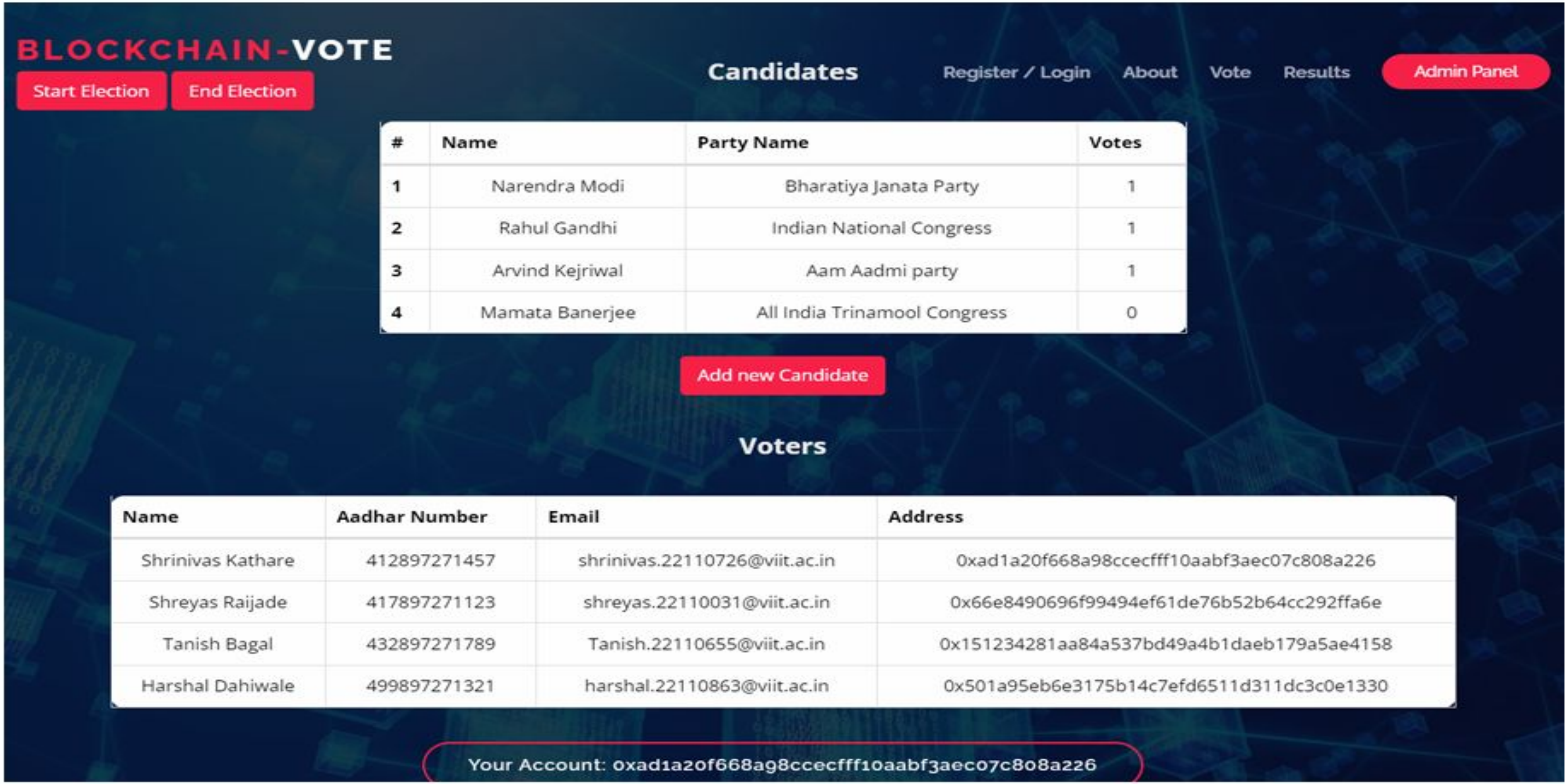
- **Vote Page**


BLOCKCHAIN-VOTE
Register / Login
About
Vote
Results
Admin Panel
Please cast your vote

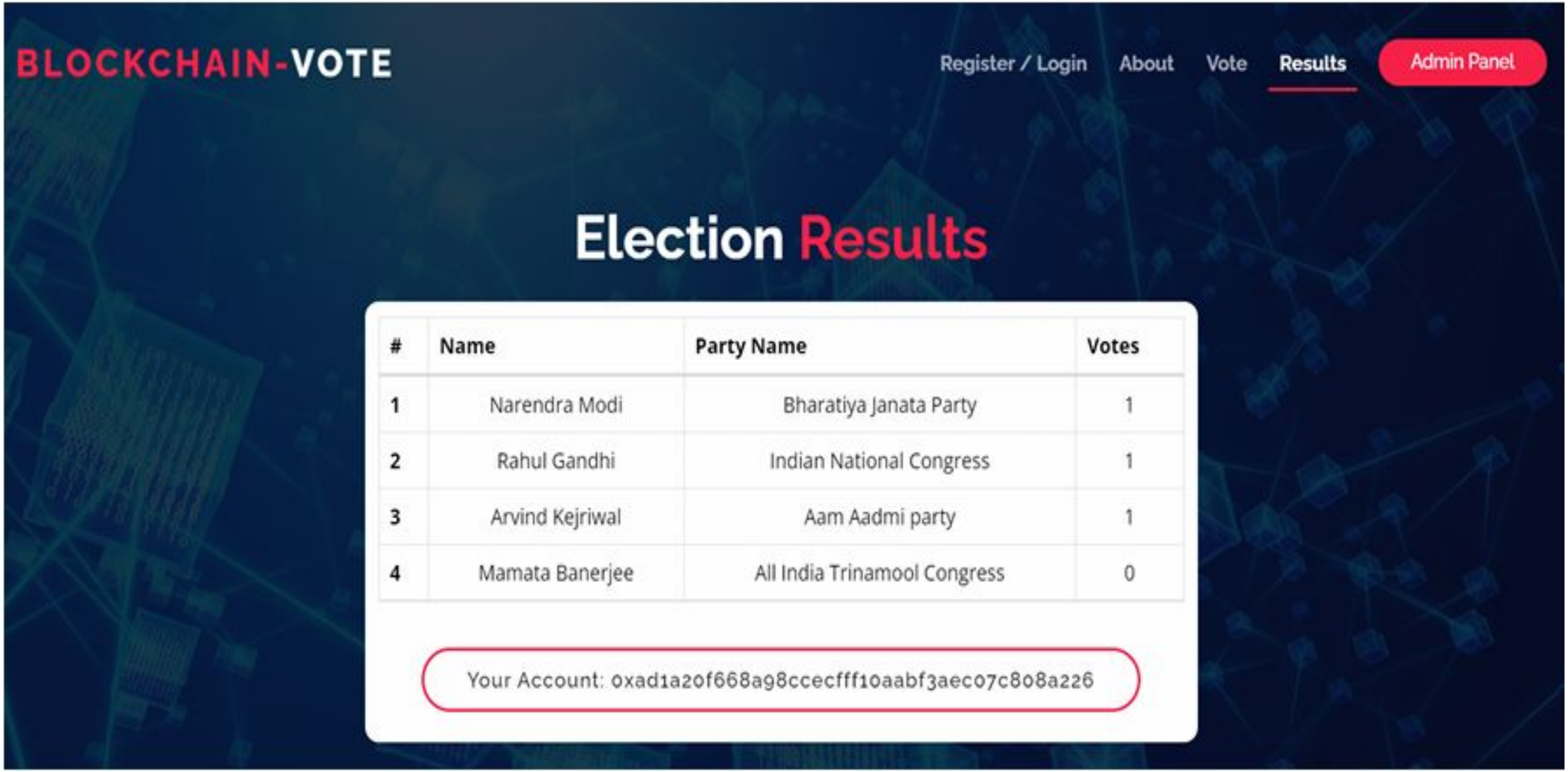
#	Name	Party Name	Votes
1	Narendra Modi	Bharatiya Janata Party	0
2	Rahul Gandhi	Indian National Congress	0
3	Arvind Kejriwal	Aam Aadmi party	0
4	Mamata Banerjee	All India Trinamool Congress	0

Select Candidate
Narendra Modi
Vote
Your Account: 0xad1a20f668a98ccecfff10aabf3aec07c808a226

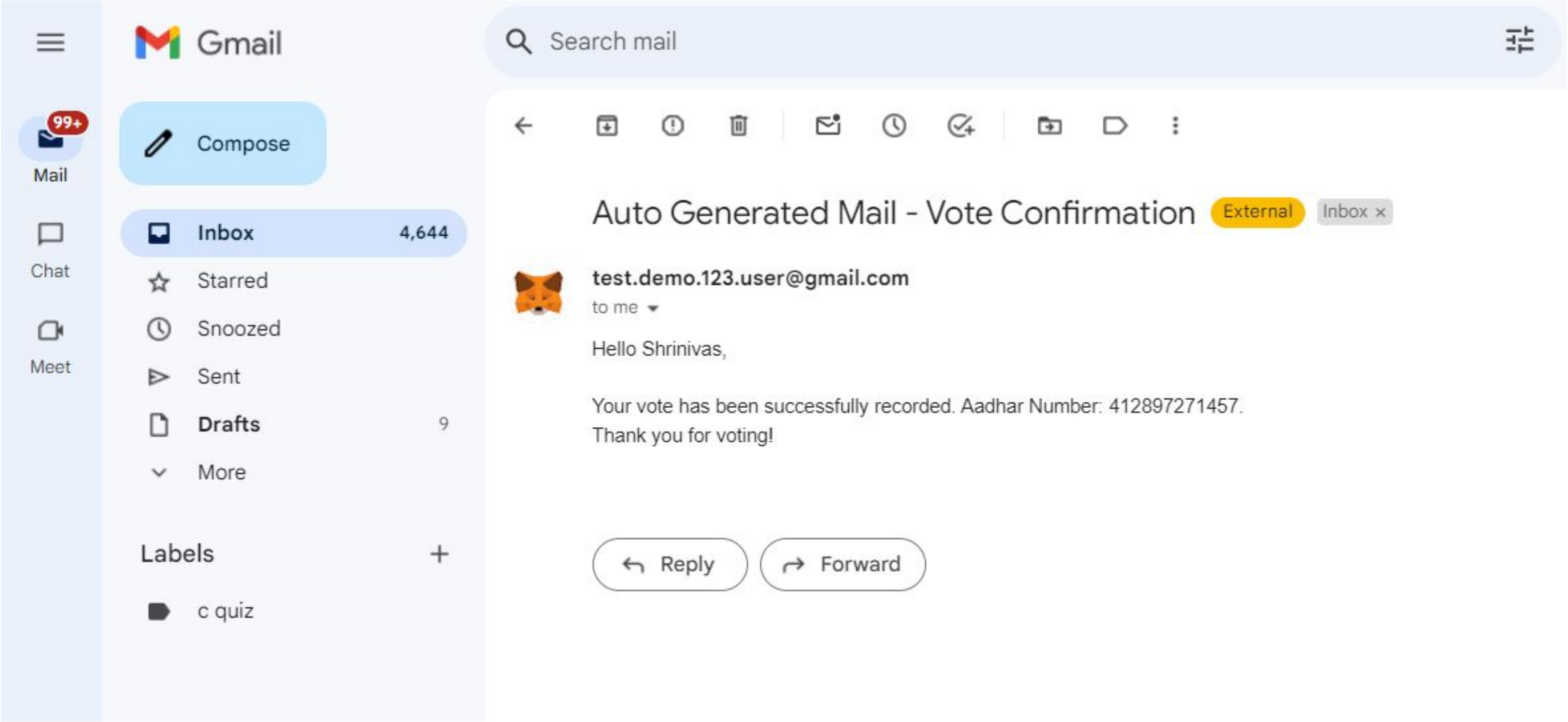
- # Admin Panel


BLOCKCHAIN-VOTE
Start Election
End Election
Candidates
Register / Login
About
Vote
Results
Admin Panel
Name Party Name Votes
1 Narendra Modi Bharatiya Janata Party 1
2 Rahul Gandhi Indian National Congress 1
3 Arvind Kejriwal Aam Aadmi party 1
4 Mamata Banerjee All India Trinamool Congress 0
Add new Candidate
Voters
Name Aadhar Number Email Address
Shrinivas Kathare 412897271457 shrinivas.22110726@viit.ac.in 0xad1a20f668a98ccecfff10aabf3aec07c808a226
Shreyas Raijade 417897271123 shreyas.22110031@viit.ac.in 0x66e8490696f99494ef61de76b52b64cc292ffa6e
Tanish Bagal 432897271789 Tanish.22110655@viit.ac.in 0x151234281aa84a537bd49a4b1daeb179a5ae4158
Harshal Dahiwale 499897271321 harshal.22110863@viit.ac.in 0x501a95eb6e3175b14c7efd6511d311dc3c0e1330
Your Account: 0xad1a20f668a98ccecfff10aabf3aec07c808a226

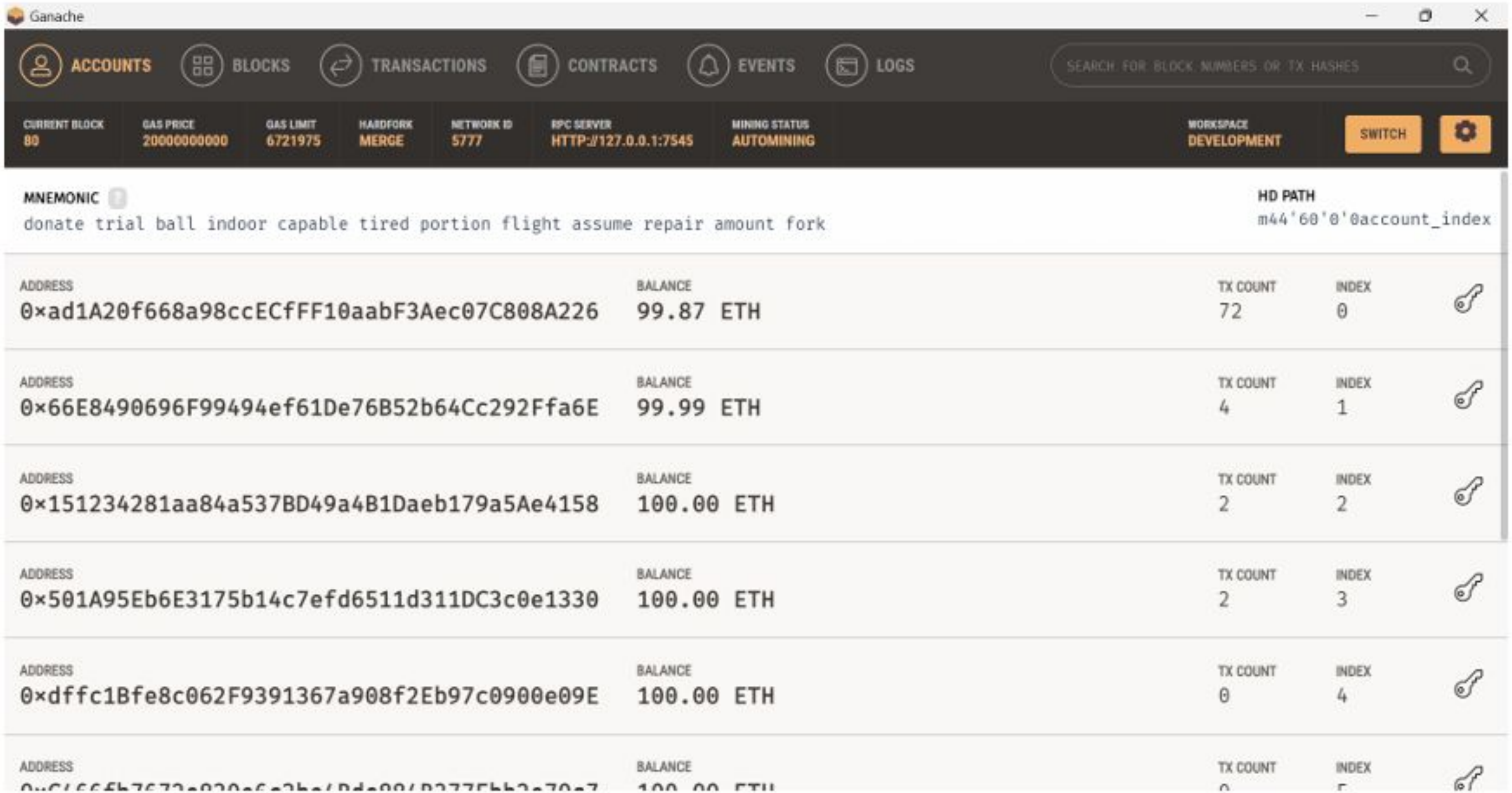
- # Result Page


BLOCKCHAIN-VOTE
Register / Login
About
Vote
Results
Admin Panel
Election Results
#
Name
Party Name
Votes
1
Narendra Modi
Bharatiya Janata Party
1
2
Rahul Gandhi
Indian National Congress
1
3
Arvind Kejriwal
Aam Aadmi party
1
4
Mamata Banerjee
All India Trinamool Congress
0
Your Account: 0xad1a20f668a98ccecfff10aabf3aec07c808a226

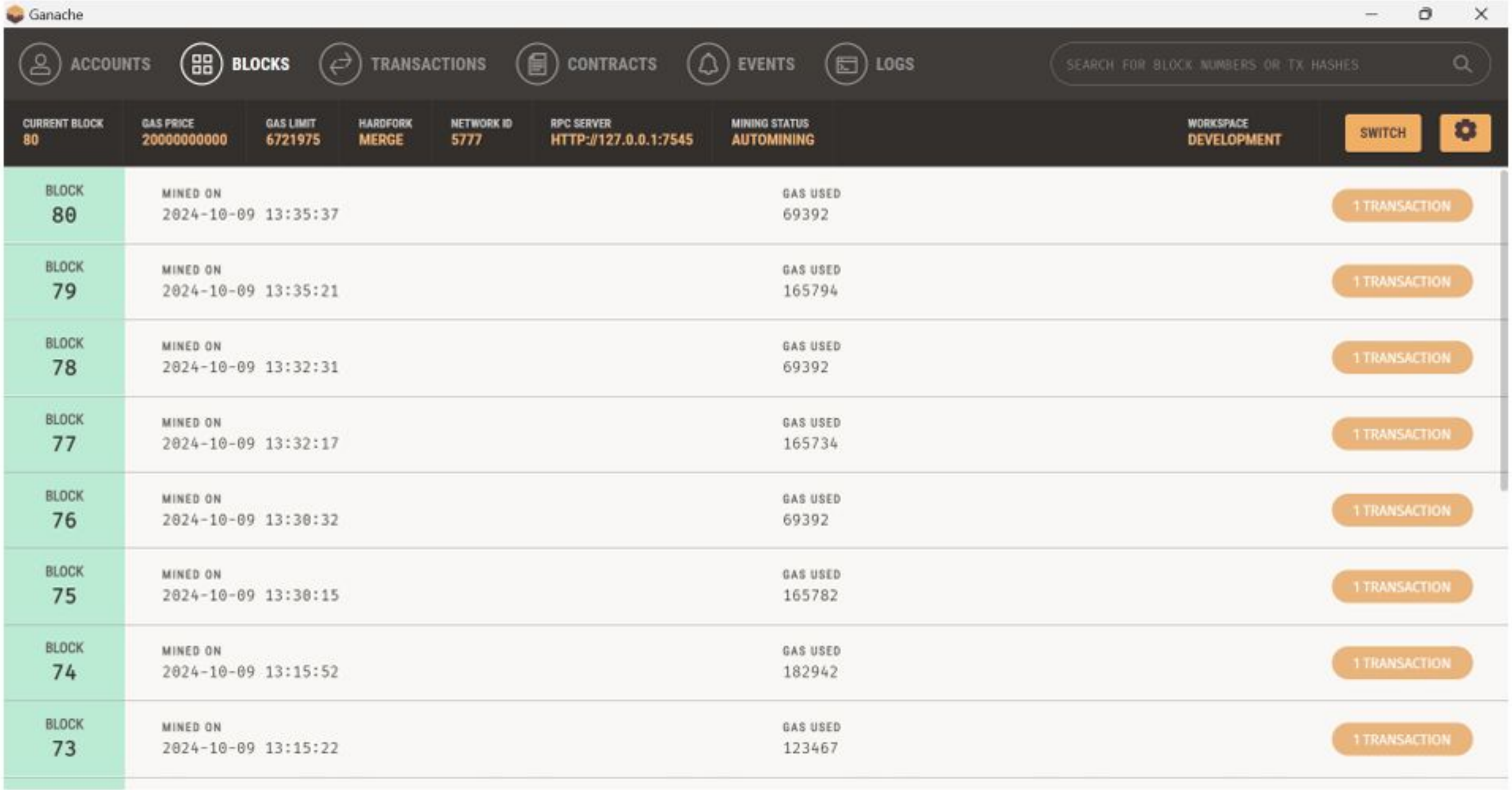
- # Vote Mail Conformation



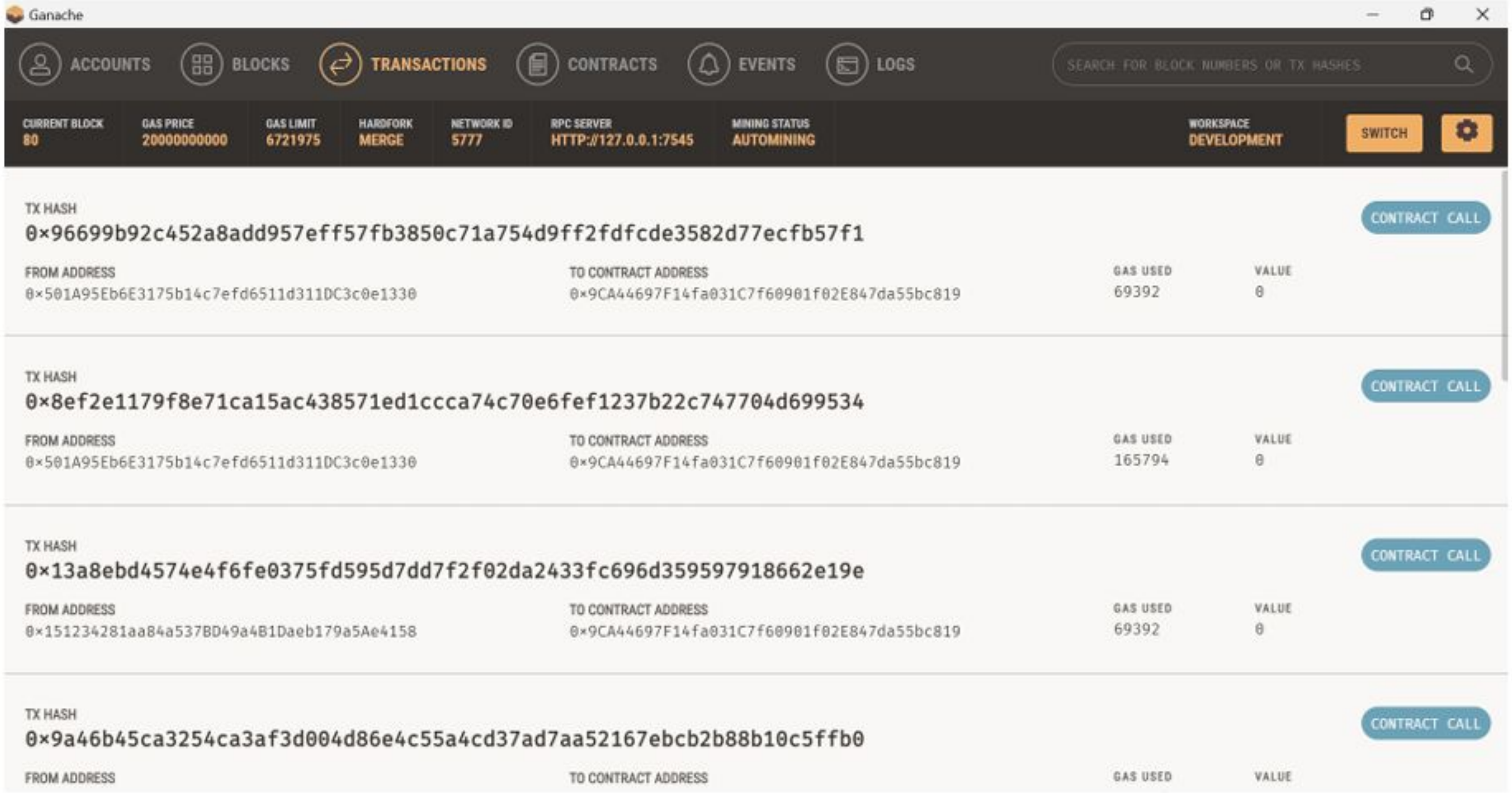
- # Ganache


Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK
80
GAS PRICE
20000000000
GAS LIMIT
6721975
HARDFORK
MERGE
NETWORK ID
5777
RPC SERVER
HTTP://127.0.0.1:7545
MINING STATUS
AUTOMINING
WORKSPACE
DEVELOPMENT
SWITCH
MNEMONIC
donate trial ball indoor capable tired portion flight assume repair amount fork
HD PATH
m44'60'0'0account_index
ADDRESS
0×ad1A20f668a98ccECfFF10aabF3Aec07C808A226
BALANCE
99.87 ETH
TX COUNT
72
INDEX
0
ADDRESS
0×66E8490696F99494ef61De76B52b64Cc292Ffa6E
BALANCE
99.99 ETH
TX COUNT
4
INDEX
1
ADDRESS
0×151234281aa84a537BD49a4B1Daeb179a5Ae4158
BALANCE
100.00 ETH
TX COUNT
2
INDEX
2
ADDRESS
0×501A95Eb6E3175b14c7efd6511d311DC3c0e1330
BALANCE
100.00 ETH
TX COUNT
2
INDEX
3
ADDRESS
0×dffc1Bfe8c062F9391367a908f2Eb97c0900e09E
BALANCE
100.00 ETH
TX COUNT
0
INDEX
4
ADDRESS
BALANCE
TX COUNT
INDEX

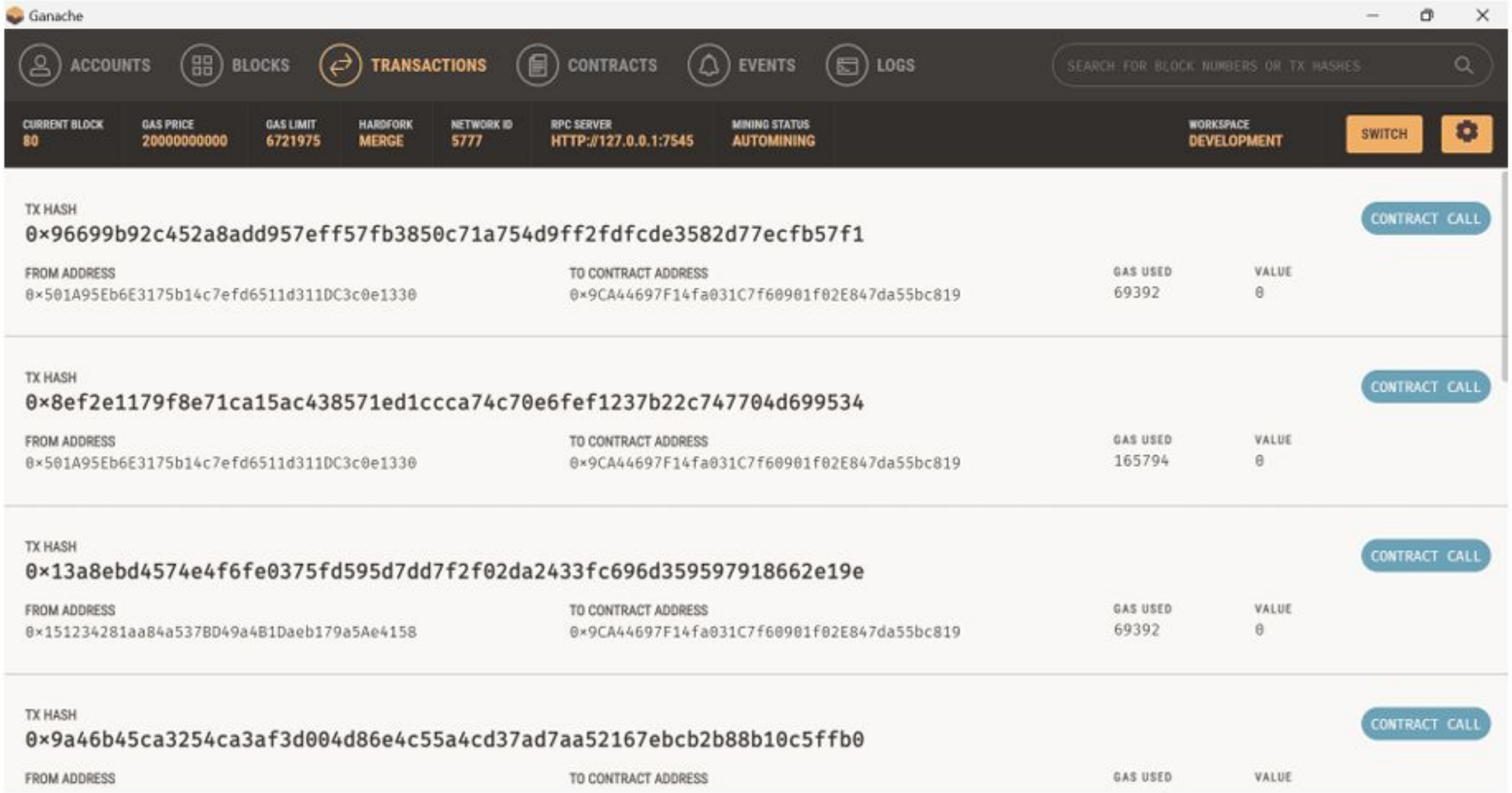
● Blocks of Blockchain



● Blockchain Transaction

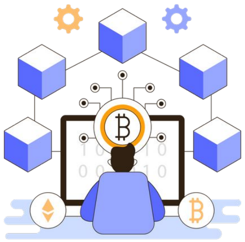


- **Blockchain Transaction**



9. Conclusion

- **Innovative Approach:** Our project leverages blockchain technology to create a decentralized, secure, and transparent voting system, aiming to revolutionize the way elections are conducted.
- **Robust Design:** By integrating frontend, backend, and blockchain components, we ensure a comprehensive solution that addresses the challenges of traditional voting systems.
- **Scalable and Versatile:** The system's architecture is designed to be scalable and adaptable, making it suitable for various types of elections, from governmental to corporate.



Thank You!